

Dimhans IDD App: Project Overview

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1. Vision & Purpose

The Dimhans IDD App is a dedicated mobile application designed to empower children with Intellectual and Developmental Disabilities (IDD). The core mission is to provide parents and caregivers with a simple, accessible, and effective tool to teach essential daily life skills.

Through a series of clear, guided video tutorials, the application helps children master fundamental tasks such as personal hygiene and self-care routines. By focusing on a visual and repetitive learning model, the app caters directly to the learning styles of its target users, fostering independence and confidence.

2. Target Audience

The primary users of this application are the parents and caregivers of children with intellectual disabilities. A special emphasis has been placed on making the app accessible to individuals in **rural and semi-urban areas**, where access to specialized therapeutic resources may be limited.

3. Technical Architecture & Design

The application is built using a modern, robust, and scalable technology stack, adhering to the principles of **Clean Architecture**. This architectural pattern separates the user interface, business logic, and data layers, resulting in an app that is highly maintainable and easy to update.

- **Platform:** Native Android
- **Language:** Kotlin
- **UI Toolkit:** Jetpack Compose (for a modern, declarative, and responsive user interface)
- **Architecture:** MVVM (Model-View-ViewModel) with Clean Architecture principles
- **Networking:** Retrofit for handling API communications.
- **Asynchronous Operations:** Kotlin Coroutines for managing background tasks smoothly.
- **Navigation:** Jetpack Navigation Compose for a seamless in-app experience.

4. Core Features

The Dimhans IDD App is equipped with a suite of features designed for security, simplicity, and effectiveness.

a. Secure & Simple Authentication: The app features a unique and highly practical authentication system. Instead of a complex, server-based database, it leverages the **Google Sheets API** as a secure and reliable backend. When a parent registers, their credentials are securely saved as a new entry in a private Google Sheet. This approach is not only cost-effective but also simplifies data management, ensuring the system is robust and easy to maintain.

b. Persistent Login & Session Management: To provide a frictionless user experience, the app uses **Jetpack DataStore** to securely save a user's login session on their device. This allows parents to open the app and be automatically logged in, without needing to enter their password every time.

c. Intuitive User Interface & Navigation: The app's design is clean and uncluttered. A simple bottom navigation bar allows users to effortlessly switch between the **Home** screen (containing the video learning modules) and a personalized **Profile** screen.

d. Personalized User Profiles: Each user has a dedicated profile that displays the child's and parents' names. This creates a personalized and welcoming environment for the family. The profile screen also includes a secure logout feature with a confirmation dialog to prevent accidental sessions ending.

e. Video-Based Learning: The core of the app is its video content. Visual demonstration is a proven, effective teaching method for children with IDD. The app provides easy access to these tutorials, which are designed to be simple and easy to follow.

f. Progress Tracking & Visualization: To help parents monitor and celebrate their child's achievements, the app includes a simple progress tracking system. Within each learning task, parents have the option to mark it as "completed." This progress is saved to the user's account, and the personalized profile screen features a dedicated section that visually displays the child's accomplishments, offering encouragement and a clear overview of the skills they have mastered.

5. Key Advantage: High Accessibility for All

A standout feature of this application is its commitment to accessibility.

The **minSdk** is set to API Level 24 (Android 7.0). This means the Dimhans IDD App is compatible with a vast majority of Android smartphones in use today, including older and more affordable models running **Android 10**. This is a deliberate and crucial design choice to ensure that the application is accessible to families in rural communities, who may not have access to the latest high-end devices. This broad compatibility directly supports the project's goal of reaching and helping as many children as possible, regardless of their family's economic status.

6. Conclusion

The Dimhans IDD App is more than just a piece of software; it is a vital resource built with care, empathy, and modern technology. By combining a thoughtful feature set with a robust and accessible technical foundation, it stands as a powerful ally for parents and a stepping stone toward greater independence for children with intellectual disabilities.